# MILITARY SPECIFICATION

## MINE, AT, PRACTICE, XM69
## PARTS FOR

### 1. SCOPE

1.1 This specification covers the parts for one type of anti-tank mine designated as Mine, AT, Practice, XM69.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9243223 - Bag, Ballast Assembly
9247199 - Cover, Protective Assembly
9254879 - Wrench, Thumbscrew
9204903 - Folding Fin Assembly
9245835 - Spool and Firing Cable Assembly

FSC: 1345

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PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET - 9256479

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials and parts.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on drawings (dwg.) 9243223, 9247199, 9254879, 9204903 and 9245835 and with all requirements specified in applicable specifications.

3.3 Spool and firing cable assembly (see drawing 9245835).

3.3.1 Continuity and resistance.-The resistance of the spool and firing cable assembly shall be eight tenths of an ohm maximum (max.) when tested as specified in 4.4.1.1.

3.3.2 Insulation resistance.-The insulation resistance between the two conductors of the cable shall be one megohm, minimum (min.) when a minimum electrical potential of 200 volts, direct current is applied to the two conductors of the cable with the shorting plug removed. The test shall be performed as specified in 4.4.1.2.

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements



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as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 complete sets of parts and 25 subassemblies (i.e., 25 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures



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and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components and subassemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 or 4.3.3 of this specification and any or all requirements of the applicable drawings. Eighteen spool and firing cable assemblies shall be tested as specified in 4.4.1.

4.2.3 Rejection.-If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or subassembly in the sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same



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materials and methods in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Fin (see dwg. 9204904 covering a detail of dwg. 9204903).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Length | Gage | 01001 |
| 102. | Width | Gage | 01002 |
| 103. | Thickness | Gage | 01003 |
| 104. | Location of holes | Gage | 01004 |
| 105. | Diameter of rivet hole | Gage | 01005 |
| 106. | Length of beveled end | Gage | 01006 |
| 107. | Radius missing | Visual | 01007 |



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Minor: AQL 0.65 percent

| | | | |
|---|---|---|---|
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 01008 |

4.3.2 2 Rivet (see dwg. 9204906 covering a detail of dwg. 9204903).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.65 percent | | |
| 101. | Outside diameter of large shank | Gage | 02001 |
| 102. | Outside diameter of small shank | Gage | 02002 |
| 103. | Length of large shank | Gage | 02003 |
| 104. | Length of small shank | Gage | 02004 |
| Minor: | AQL 1.0 percent | | |
| 201. | Inside diameter missing | Visual | 02005 |
| 202. | Evidence of poor workmanship (see 3.4) | Visual | 02006 |

4.3.2.3 Band, fin, folding (see dwg. 9204907 covering a detail of dwg. 9204903).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 1.0 percent | | |
| 101. | Thickness of material, min | Gage | 03001 |
| 102. | Height of fin mounting plate | Gage | 03002 |
| 103. | Location of holes | Gage | 03003 |
| 104. | Inside diameter of band | Gage | 03004 |
| 105. | Location of fin mounting plate | Gage | 03005 |
| Minor: | AQL 1.5 percent | | |
| 201. | Width of band | Gage | 03006 |
| 202. | Width of fin mounting plate | Gage | 03007 |
| 203. | Evidence of poor wormanship (see 3.4) | Visual | 03008 |



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4.3.2.4 Folding fin assembly (see dwg. 9204903).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Rivet loose | Manual | 04001 |
| 102. | Fins do not spring back to fully extended position | Visual-Manual | 04002 |
| 103. | Operation missing or improperly performed | Visual | 04003 |
| 104. | Component missing or improperly assembled | Visual | 04004 |
| 105. | Load required to lift loop | Test | 04005 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 04005 |

4.3.2.5 Firing cable assembly (see dwg. 9235372 covering a detail of dwg. 9245835).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.0 percent | | |
| 101. | Length | Gage | 05001 |
| 102. | Component missing or improperly assembled | Visual | 05002 |
| 103. | Potting compound omitted | Visual | 05003 |
| 104. | Wiring missing, broken or incorrect | Visual | 05004 |
| 105. | Marking missing, incorrect or unidentifiable | Visual | 05005 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 05006 |

4.3.2.6 Spool and Firing Cable Assembly (see dwg. 9245835).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |



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Major: AQL 0.40 percent
101. Component missing or improperly assembled.....Visual 06001

Minor: AQL 0.65 percent
201. Evidence of poor workmanship (see 3.4).....Visual 06002

4.3.2.7 Bag, ballast assembly (see dwg. 9243223).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: None defined | | | |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 07001 |

4.3.2.8 Cover, protective assembly (see dwg. 9247199).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: None defined. | | | |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 08001 |

4.3.2.9 Wrench, thumbscrew (see dwg. 9254879).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: None defined. | | | |
| Major: | AQL 0.40 percent | | |
| 101. | Width of slots | Gage | 09001 |
| Minor: | AQL 0.65 percent | | |
| 201. | Evidence of poor workmanship (see 3.4) | Visual | 09002 |



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4.3.3 Testing

4.3.3.1 Spool and firing cable assembly

4.3.3.1.1 Continuity and resistance (see 3.3.1) - Major defect - Code No. 10001.-This test shall be performed one hundred percent. Any cable which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.1.1 using equipment in accordance with 4.3.4.

4.3.3.1.2 Insulation resistance (see 3.3.2) - Major defect - Code No. 11001.-This test shall be performed one hundred percent. Any cable which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.1.2 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-9256479 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Spool and firing cable assembly

4.4.1.1 Continuity and resistance.-The resistance reading shall be taken at 70 plus or minus 20 degrees Fahrenheit (F) at the rocket and terminals of the cable with the shorting plug assembled to the male contacts. Observation shall be made to determine compliance with the applicable requirements. Test equipment shall be in accordance with 4.3.4.



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4.4.1.2 Insulation resistance.-The test for insulation resistance shall be performed by applying the specified electrical potential to the cable with the shorting plug removed. Observation shall be made to determine compliance with the applicable requirement. Test equipment shall be in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Level C.-Inert component parts shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.



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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.



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6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design reveiw will be based on the date of the final submission of designs. Picatinny Arsenal design reveiw will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Prject Number: 1345-A-310



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